



FuzzEG: Fuzzy logic for adaptive scenarios in an educational adventure game

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Abstract

Nowadays, there is an increased interest in using computer games for educational purposes. Educational computer games attempt to increase the students' motivation and engagement. However, they should provide adaptivity to learners' different needs and characteristics for better educational results. Towards this direction, this paper presents a novel adaptive educational game named FuzzEG, which teaches the knowledge domain of HTML programming language. It uses fuzzy sets to represent the learner's knowledge level more realistically. Taking into consideration the learner's progress, the game decides about the level of difficulty of the quizzes and whether the scenario is going to be dynamically expanded or not. The gain of this is that it provides the learner with a personalised learning and gaming experience. The game has been evaluated, and the results are very encouraging.

Keywords Fuzzy logic · Edutainment · Adventure games · Adaptive games

1 Introduction

Computer games are a popular way of entertainment. Nowadays, more and more people of different ages play digital games (usually web or mobile games) in their leisure time. Computer games' interactive content, which includes videos, animations, texts, and audio, make them exciting and stimulating. As a result, they cause enthusiasm and pleasant feelings to users keeping them engaged. That is the reason for the tendency to use them for educational purposes. The computer games technology could be used to render educational software more motivating and engaging, making the painful process of learning amusing.

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Many researchers have pointed out the positive impact of computer games in education. Particularly, Seymour Papert, who is internationally recognized as the seminal thinker regarding computers and pedagogy for children, notes that software games teach children that some forms of learning are fast-paced, immensely compelling and rewarding whereas, by comparison, school strikes many young people as slow and boring [16]. Also, Boyle points out that games can produce engagement and delight in learning; they thus offer a persuasive format for educational environments [1]. Furthermore, traditional learning methodologies are not appealing students to study anymore while educational video games encourage students to learn in new ways or enjoy otherwise tedious tasks [12]. Therefore, digital games are ideal to be used for educational purposes.

Games are associated with one or more categories according to their style. For example, in platform games, the hero jumps from platform to platform while in shooter games the player faces enemies from a distance using ranged weapons. Instead, strategy games require careful thinking, planning and creating tactics to help the player achieve her/his goal. Some games can combine items from two or more categories. A question that has arisen is which types are suitable for use in education. Researchers have shown that the category of adventure games contains elements that are beneficial to the learning process [13, 22, 23]. These are games where the player assumes the role of a protagonist in a story that evolves based on exploration and puzzle-solving [4]. The narrative that accompanies an adventure game can be used as a means of transferring the educational content to the user. The absence of a time limit helps the student to train at her/his own pace and to understand the subject matter better [19].

However, an educational game has to provide adaptivity to learners' different needs and characteristics for better educational results. The educational game has to be able to readjust each time the instructional process and the teaching strategy considering the student's needs and abilities, as a teacher does in a real-life classroom. Consequently, creating an adaptive learning system that meets the students' requirements can be challenging since students learn with not only different needs but also different learning characteristics [11]. A solution for this is the fuzzy logic since it has been introduced by Zadeh [28] to handle uncertainty and imprecise data. In particular, learning is a complicated process, and knowledge level is a notion that cannot be quantified as a variable which accepts distinct values. Therefore, fuzzy sets and fuzzy logic can be used to represent the learner's knowledge level more realistically. Fuzzy logic techniques can be used to improve the performance of an educational environment since they can encounter the uncertainty that there is in information about students' cognitive state and behaviour.

The main goal of the research described in this paper is to present an educational adventure game that helps users to learn HTML programming. The game uses all the elements contained in a typical adventure game. The hero can walk to several rooms, collect items, combine some of them, and talk with different non-playable characters (NPCs). The ultimate goal is to escape from the building where he is imprisoned. In some rooms, some characters offer to help the hero as long he answers various quiz questions. It is a prerequisite for the hero to respond successfully to a good percentage of questions to receive a reward useful for the evolution of the story.

The representation of the learner's knowledge level is realised through fuzzy sets each time. Then, the game decides the quizzes and the game's flow. In particular, the game takes into consideration, each time the learner's progress, and decides about the quizzes that have to be delivered to the learner dynamically. In particular, it decides either about the number of quizzes that s/he has to answer, and consequently, about the rooms from which s/he has to escape to finish the game or the number and difficulty level of the questions that consist

the quiz each time. Therefore, the student is offered a more personalised instruction using the fuzzy logic techniques which take as input the results in the quizzes and decides about the questions that are suitable to be delivered to the learner to the next quizzes of the game. Simultaneously, the student/player is offered a more personalised gaming experience, since the rooms from which s/he has to escape to finish the game, are chosen to be displayed, dynamically, according to the student's progress and needs. The game has been evaluated, and the results are very encouraging.

The remainder of this paper is organized as follows. Section 2 presents the related work in games and adaptivity. Section 3 describes the game. Section 4 describes how adaptivity is realized by using fuzzy sets. Section 5 describes examples of the operation of the game. Section 6 presents experimental data derived from the game's use. Section 7 depicts the results of the game's evaluation. Finally, in Section 8, conclusions are drawn.

2 Review

One of the main reasons people use computers is for playing games. Digital games offer fun, entertainment, challenge, and relaxation. Games are a category of computer applications which exist for several years. In 1970 and 1980, video games were introduced to the general public. Since then, video gaming has become a popular form of entertainment. The impact that digital games had, resulted in their rapid development into the massive entertainment industry [15]. Their primary purpose is to motivate users and keep them engaged in the game by creating a pleasant feel to them.

Video games can be divided into several categories, called genres. Some combine different gaming styles and, thus, can appear under more than one category. Moreover, some genres can be divided into subcategories. Shooter games are games that require the player to blow away enemies or objects to survive and continue game play. Sports games emulate the playing of traditional physical sports. Some emphasise playing the sport, while others emphasise the strategy behind the sport [26]. Fighting games simulate close-range combat against a few opponents. While ranged and melee weapons may be present in fighting games, they emphasise hand-to-hand combat.

One of the first video game genres was the category of adventure games, which were initially text-based. Nevertheless, as the power of the gaming system and technology grew, the graphic adventure format became popular moving towards point and click interfaces. They are not based on what the dictionary defines as "adventure", but the name stems from the game "Adventure", which is also known as Colossal Cave Adventure [27].

Also, educational games constitute another genre, that becomes increasingly popular, as game-based learning has become a research trend in the field of educational technology. The spreading recognition of games in leisure time among people of different ages [6] has led to educators' conjecture that educational games should also be included in learning contexts to engage students [22]. Their main objective is to try to teach the users through playing. They are designed as fun activities through which students can learn and practice different concepts [21]. The outcomes of their use in the learning process have been examined and discussed by a large number of researchers [8].

As stated earlier, a video game may belong to more than one category. Given this, a question that arises is which game category is suitable and useful for learning. V. Riemer and C. Schrader argued that three advantageous game categories for learning are quizzes, simulations and adventure games [22]. The narrative framework that accompanies adventure games can fit more comfortably to learning needs as they emphasise on learning instead

of action [19]. Educational content can be embedded in the game world utilising the game design and players can take their time in absorbing the presented knowledge due to the absence of time limits [13]. Adventure games use the intrinsic motivation of the player to explore the game world. Intrinsically motivating games may incorporate learning activities into their game world [20].

Despite the above finding, there is a lack of educational games since there are only a few implementations. C. D'Apice et al. developed an educational adventure game, called SIRET Security Game, which aims to teach workers about computer security policies [4]. The player takes the role of an employee of big business, and her/his objective is to protect sensitive company data from potential spies and hackers. H. Hou and M. Li developed Boom Room, a problem-solving-based adventure, in which the student has to collect hardware and assemble a desktop computer to disable a bomb and escape a room within 10 minutes [8]. The evaluation of the game indicated that participating students perceived the game to be useful for their learning. Gaeta M. et al. created ALICE Project, whose main feature is its narrative style [7]. It provides a simple storytelling interface, but it does not contain the rest of the elements that characterize a typical adventure game.

Web-Based tutoring or learning application is used by a very diverse audience, which consists of learners with different characteristics and needs [3]. This fact emphasises the need to develop web-based educational systems that can offer dynamic adaptation to each student. Considering this ascertainment, researchers have proposed encompassing adaptive features to the development of educational games as well [17, 29]. Adaptivity in game-based learning aims to support and encourage the player by taking into account the player's possible weaknesses. An adaptive game may be used to examine and determine the students' knowledge level. It can identify students' possible difficulties trying to solve specific problem types and help them overcome them. The game can adapt to students' needs and provide valid assessment information to students and teachers as it suits to individual players while they play to entertain and learn more effectively [10]. It may collect various information about their moves or answers and make the necessary adjustments to provide the student with the correct guides or additional help.

Despite adaptivity's importance and positive influence on students, there are only a few examples of educational games that encompass it. SIRET Security Game mentioned above calculates the player's number of failed attempts or the time spent trying to solve a puzzle. Depending on the calculations, the player may receive a phone call from her/his mentor who provides instructions and tips for solving the quests. On the contrary, Boom Room does not contain adaptive characteristics. However, the creators suggest that the design of Boom Room and other similar games can be significantly enhanced by providing tasks with varying levels of difficulty depending on the player's progress. ALICE Project embodies more adaptive features than the previous examples. The story is adapted to the needs of the user by offering a different perspective and point of view. The game takes advantage of a user modelling component which calculates the player's success rate in a test.

There is only a small number of educational adaptive adventure games, and their evaluation of players' knowledge level is based on simple and basic models or calculations. Because of this, an educational adventure game that is based on a different model to assess the student's cognitive state and adapts appropriately could be beneficial. Learning is a complicated process, and the knowledge level is a notion that cannot be quantified. Knowledge cannot be considered as a variable which accepts distinct values. That is, it is not precise to say that a domain concept is unknown, known or assimilated. Problems that include entities which cannot be measured can be addressed by fuzzy logic [28].

As a consequence, fuzzy logic seems to be an ideal methodology for representing the knowledge level. It has been used in many adaptive tutoring systems [23–30], but not in educational games. Taking into consideration this fact, we present an educational adventure game that incorporates a novel mechanism that uses fuzzy logic to build student models and adapt the quizzes of the game and alter the plot of the game on the fly.

3 Game description

In “HTML Escape Game”, the player takes the role of a hero who is trapped and locked in a building. The world in which the hero moves is a house with several rooms and a garden. In the first room, there is a screen on the wall, where a character appears and informs the hero about her/his imprisonment. The game follows the principles of a typical adventure game whose story is driven by exploration and puzzle solving. The main character has to complete a series of riddles and small quests to find the exit and escape. The hero may move to different rooms, talk to other non-playable characters (NPCs) and gather and use items. Solving a mini-quest unlocks new riddles, rooms, items or characters.

The game makes use of an inventory system where the hero keeps the gathered objects (Fig. 1). Only specific objects can be picked up in the game, so the player knows that these objects are essential for the continuation of the story and the escape attempt. For example, by putting a piece of lemon in a glass of soda, the hero can create a tasty and refreshing drink and, subsequently, give it to a thirsty NPC. Afterwards, the NPC rewards the player with an item that is critical for continuing the game.

Figure 2 presents a diagram of some of the locations where the hero can walk and the actions that can be taken in each of them. In some locations in the diagram, there is a “quiz” indication meaning that on that screen the player is asked to answer quiz questions. For example, at the “gym” location, the player must complete the quiz so that the NPC gives her/him an oil-can full of oil.

Figure 3 shows a diagram of the actions the hero must perform to escape. Ellipses depict actions the hero can perform without the need for another action (e.g. “Take glow stick”,



Fig. 1 The main game environment

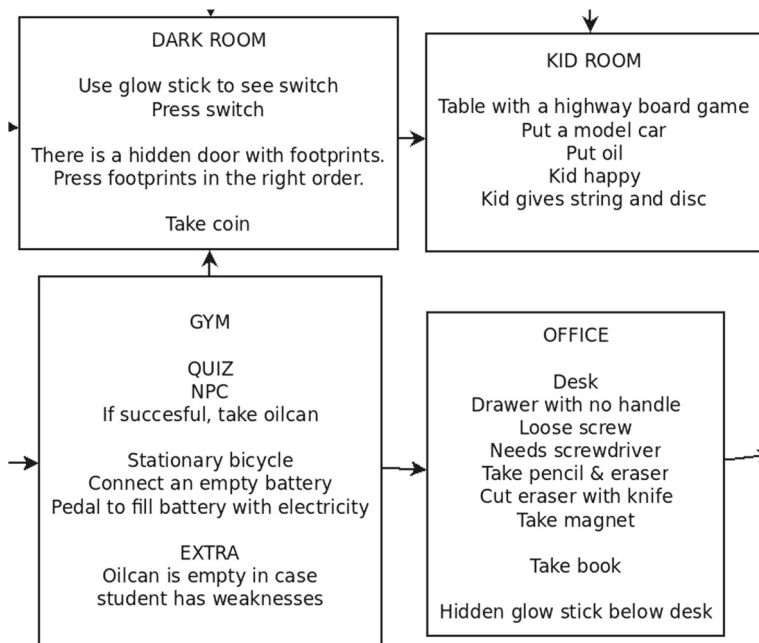


Fig. 2 Diagram of some game rooms and locations

“Take battery”). In contrast, green rectangles are actions that depend on others. For the sake of clarity of the sample, the image does not show the complete action diagram and some of the dependencies of the projected actions (e.g. “Connect the hose to magic hydra”, “Place letter E”). The final action (“ESCAPE”) is the rectangle with the brown colour. According to the chart, the hero should get the glow stick that helps him dimly illuminate a dark room. Later he observes that there is a switch and he can push it to turn on the light. With the light on, he sees that there are wall prints that lead to a secret passage. After solving the puzzle, the hero finds a tile with the letter F. Placing the tile F (along with tiles I, R and E) into individual slots leads to a new secret passage and a step before the escape.

The tree does not include the quizzes, but only the actions that are related to the game context. The tree structure of the actions was used to examine and confirm that no circle of actions was created. Circle of actions would mean that the hero could enter a state from which he cannot come out and, thus, it is impossible to complete the game. Impossible completion of the game may have resulted in the player’s disappointment or frustration and the lack of motivation to re-engage with the game.

Additionally, the full hierarchy tree shows that it is necessary to execute all the actions and solve all the individual puzzles to complete the game. During the puzzle making process, it was noticed that the player could complete the game in only three steps. Using the actions chart, we ensured that all steps must be taken to finish the game. Possible completion of the game with minimal steps could potentially reduce the player’s motivation.

As the title states, the learning goal of the game is HTML programming language. To integrate educational features into the game, we extended the game design by adding quizzes at several points of the game. These quizzes are related to specific concepts of HTML. They are distinguished in primary and extra. It is obligatory to solve the primary ones so



Fig. 3 Sample of actions needed to be taken

that the story of the game proceeds. The extra ones arise under particular circumstances. In these cases, it is necessary to resolve them as well. When a player resolves a quiz with a satisfactory degree of success, the game may unlock new elements and riddles. The selection of the quiz format is based on the fact that it is one of the simplest and most popular ways to test [14]. A quiz may include questions of the following types: single choice (e.g. True or False), multiple choice, sorting and filling spaces in a text. Each question is characterized by a difficulty level (easy, moderate, hard) and may have two or more candidate answers as there is no maximum limit.

The implementation of the game is based on web technologies. Given this, the game is accessible through any computer that has access to the Internet. The application is separated into two distinct yet interrelated components (Fig. 4): the main game component and the educational component. The first one includes the game context, where the player moves and interacts with characters and objects. It can be detached from the educational component and can be used as a typical adventure game. However, if the game is used for educational purposes, when a quiz is about to show up, the educational tool feeds the game component with the necessary information. The educational tool includes the questions about a chapter, the corresponding possible answers, and the question type. The educational tool examines the correctness of each answer and informs the user accordingly. In case of incorrect answers, there is no feedback regarding the correct one or ones (in case of multiple-choice questions). The communication between the two components is achieved through web services. The educational component is the producer side which provides essential methods and

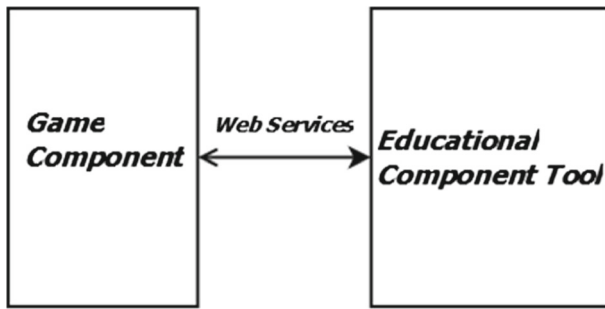


Fig. 4 Main components of the System

functionality while the game component consumes these methods. For example, there are methods for creating new quizzes about a user and specific chapters, checking the answers of the user or retrieving her/his estimated knowledge level. Concerning the latter, the game incorporates adaptive features that can affect it to a small or large extent. Embedding adaptivity in the game increases the player's intrinsic motivation and keeps her/him occupied with it more time.

Additionally, to add HTML material and exercises, we used an authoring tool that we developed for this purpose. The authoring tool enables any teacher to add educational content on any topic without having any programming skills. At the start of the game, the player selects the learning object s/he desires. For the needs of this project, the student can only select HTML, hence the title "HTML Escape Game". Finally, the description of the authoring tool is out of the scope of this article.

4 Adaptivity with fuzzy logic

Determining the student's knowledge is not a straightforward task since it often depends on and is reflected through things that cannot be directly observed and measured [9]. It is not precise to say that a domain concept is learned or unknown, since a part of it may be unsatisfactorily known, and another part may be well known. Therefore, student's knowledge cannot be considered as a variable which takes discrete values, since its determination deals with uncertainty and human subjectivity. One possible approach to encounter this is fuzzy logic, which was introduced by Zadeh [28], to handle uncertainty in everyday problems caused by imprecise and incomplete data, as well as human subjectivity [5].

We define the following four fuzzy sets for describing the student knowledge of a domain concept (Fig. 5), based on previous work of Chrysafiadi [2]:

- **Unknown (Un):** the student does not know the domain concept. The degree of success in the domain concept's quiz is from 0% to 50%.
- **Unsatisfactorily Known (UK):** the student has little or moderate knowledge of the domain concept. The degree of success in the domain concept's quiz is from 45% to 70%.
- **Known (K):** the student has good knowledge about the domain concept. The degree of success in the domain concept's quiz is from 65% to 90%.
- **Learned (L):** the student has learned very well the domain concept. The degree of success in the domain concept's quiz is from 85% to 100%.

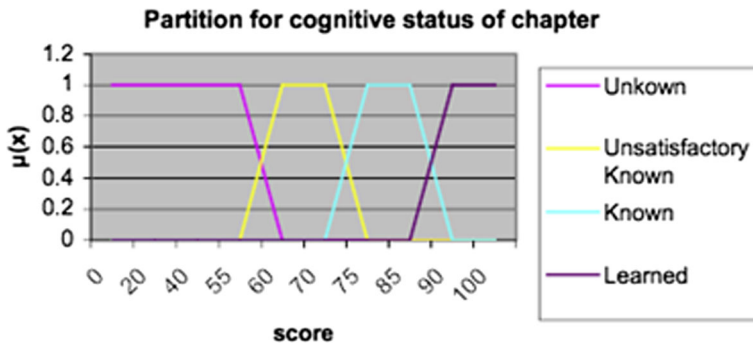


Fig. 5 Partition of the cognitive status of the chapter

The membership functions for the four fuzzy sets are defined below. Membership functions take the student's success rate (x) in a quiz as input and calculate the degree of truth in each fuzzy set.

$$\mu_{Un}(x) = \begin{cases} 1, & x < 45 \\ 1 - \frac{x-45}{5}, & 45 < x < 50 \\ 0, & x \geq 50 \end{cases} \quad (1)$$

$$\mu_{UK}(x) = \begin{cases} \frac{x-45}{5}, & 45 < x < 50 \\ 1, & 50 \leq x \leq 65 \\ 1 - \frac{x-65}{5}, & 65 < x < 70 \\ 0, & x \leq 45, x \geq 70 \end{cases} \quad (2)$$

$$\mu_K(x) = \begin{cases} \frac{x-65}{5}, & 65 < x < 70 \\ 1, & 70 \leq x \leq 85 \\ 1 - \frac{x-85}{5}, & 85 < x < 90 \\ 0, & x \leq 65, x \geq 90 \end{cases} \quad (3)$$

$$\mu_L(x) = \begin{cases} \frac{x-85}{5}, & 85 < x < 90 \\ 1, & 90 \leq x \leq 100 \\ 0, & x \leq 85 \end{cases} \quad (4)$$

The thresholds of the above fuzzy sets have been defined by experience and they can be changed according to the teaching needs and requirements of the instructor each time. The above membership functions can take values from 0 to 1, that is:

$$\mu_{Un}(x), \mu_{UK}(x), \mu_K(x), \mu_L(x) \in [0, 1]$$

Furthermore, there is the limitation that the different knowledge levels of a domain concept can be only of adjoining fuzzy sets. Mainly, if z is a domain concept, its possible cognitive states based on the defined fuzzy sets could be the following ones:

- The concept is unknown.
- A part of the concept is unknown, and the rest of it is unsatisfactorily known.
- The concept is unsatisfactorily known.
- A part of the concept is unsatisfactorily known, and the rest of it is known.
- The concept is known.
- A part of the concept is known, and the rest of it is learned.
- The concept is learned.

Moreover, for a more natural way of representation, the value of the knowledge level that corresponds totally to a domain concept is considered to be 1, which declares a percentage of 100% knowledge. Therefore, the sum of the partition value of each fuzzy set that represents the knowledge level of a domain concept each time equals to 1. So, the relation $\mu_{U_n}(x) + \mu_{UK}(x) + \mu_K(x) + \mu_L(x) = 1$ stands. Below, some examples of the possible knowledge level of the domain concept z are described:

- If $\mu_{U_n}(x) = 1$, then it means that the domain concept z is 100% unknown.
- If $\mu_{UK}(x) = 0.3$ and $\mu_K(x) = 0.7$, then it means that the domain concept z is 30% unsatisfactorily known and 70% known ($\mu_{UK}(x) + \mu_K(x) = 0.3 + 0.7 = 1$).
- If $\mu_K(x) = 0.8$ and $\mu_L(x) = 0.2$, then it means that the domain concept z is 80% known and 20% learned ($\mu_K(x) + \mu_L(x) = 0.8 + 0.2 = 1$).
- If $\mu_L(x) = 1$, then it means that the domain concept z is 100% learned.

When the player completes a quiz, the student modelling component is fed with the corresponding success rate, calculates the degree of membership in each fuzzy set and informs both the educational and the main game component accordingly.

During the process of selecting questions, the educational tool exploits the information provided by the student modelling component. The first time the user is asked to solve a quiz, there is no information about her/his classification in one or more of the above-mentioned fuzzy sets. In this case, the system randomly selects 15 easy questions. When the player completes the quiz, the success rate is the input to the membership functions. In this way, the player is assigned to one or two (adjacent) fuzzy sets and the corresponding knowledge level is calculated. At the beginning of the second quiz, the choice of questions depends on the knowledge level of the player defined in the previous interaction. After completing the second quiz, the student's level of knowledge is re-calculated. From this point on, in each subsequent quiz, the choice of questions is based on the player's rankings in the previous two quizzes. Let Y the knowledge level determined by the user's performance in the previous quiz and X the knowledge level estimated in the quiz right before. Table 1 contains the rules for choosing quiz questions.

The student's knowledge level directly affects the level of difficulty of a quiz and the number of queries that constitute it. The quiz changes dynamically depending on the user's answers to cover any deficiencies but also continually improves her/his level of knowledge with the ultimate goal of fully assimilating the concept of HTML programming.

However, apart from the educational part, adaptivity applies to the game context itself. Game adaptivity is also based on simple linguistic rules that simulate human logic and may affect the game's flow. These rules apply when the player completes a quiz, and the game calculates the player's knowledge level:

1. If the knowledge level is "Unknown" or "Unsatisfactorily known", the user has failed the quiz, and there is no reward. The player has to talk again to the NPC that assigned the quiz to her/him and try to solve a new quiz containing questions tailored to the player's knowledge level.
2. If the knowledge level is "Learned", the user completes the quiz successfully and takes a reward that is essential to continue the game.
3. If the quiz belongs to the primary ones and the knowledge level is "Known" but not "Learned", the user has completed the quiz. However, in this case, the game assumes that the player has some weaknesses regarding the chapter s/he was examined. The game does not show any relevant message to inform the player but gives her/him a reward which is smaller compared to the one in rule 2. At the same time, a seamless

Table 1 Rules that affect the quizzes

Quiz	X	Y	Number of questions			Rule number
			Easy	Moderate	Hard	
1st	Not applicable	Not applicable	15	0	0	1
2nd	Not applicable	$X = Un, \mu_{Un} \in [0, 1]$ or	15	0	0	2
		$X = UK, \mu_{UK} < 1$				
		$X = UK, \mu_{UK} = 1$	7	8	0	3
		$X = K, \mu_K < 1$	—	15	—	4
		$X = K, \mu_K = 1$	—	7	8	5
		$X = L, \mu_L \in [0, 1]$	—	—	15	6
		$Un, \mu_{Un} \in [0, 1]$ or	15	—	—	7
		$UK, \mu_{UK} \in [0, 1]$				
		$UK, \mu_K < 1$				
		$UK, \mu_K = 1$	7	8	—	8
		$K, \mu_K < 1$	—	15	—	9
		$K, \mu_K = 1$	—	7	8	10
		$L, \mu_L \in [0, 1]$	—	—	15	11
		$Un, \mu_{Un} \in [0, 1]$ or	7	8		12
		$UK, \mu_{UK} \in [0, 1]$				
		$K, \mu_K < 1$	—	15	—	13
		$K, \mu_K = 1$	—	7	8	14
		$L, \mu_L \in [0, 1]$	—	—	15	15
3rd, 4th, 5th,...	$K, \mu_K \in [0, 1]$	$Un, \mu_{Un} \in [0, 1]$ or	7	8	—	16
		$UK, \mu_{UK} \in [0, 1]$				
		$K, \mu_K < 1$	3	8	4	17
		$K, \mu_K = 1$	—	7	8	18
		$L, \mu_K = 1$	—	—	15	19

- process runs in the background which expands the game scenario by adding new rooms, characters, items and quizzes (Fig. 6). The player has to solve the additional riddles to continue the game and fulfil her/his final quest.
4. If the quiz is extra one and the knowledge level is anything but “Learned, the user has failed and must repeat it until the exported model includes the fuzzy set “Learned”.

Given the above, the selection of questions follows the rules that are reflected above. Furthermore, the player has to reach the “Learned” status for each chapter s/he is examined to complete the game.

5 Examples of operation

A group of 60 users used the presented educational game. All the users were learners of a postgraduate program in the field of informatics at the University of Piraeus. To illustrate how fuzzy logic affects both the learning process object and the game itself, we describe some representative examples of use. Each case consists of a series of quizzes that the player tries to solve. Each interaction with a quiz is divided into three sub-elements: description, educational part, and game part. In the description, we mention some action of the player,

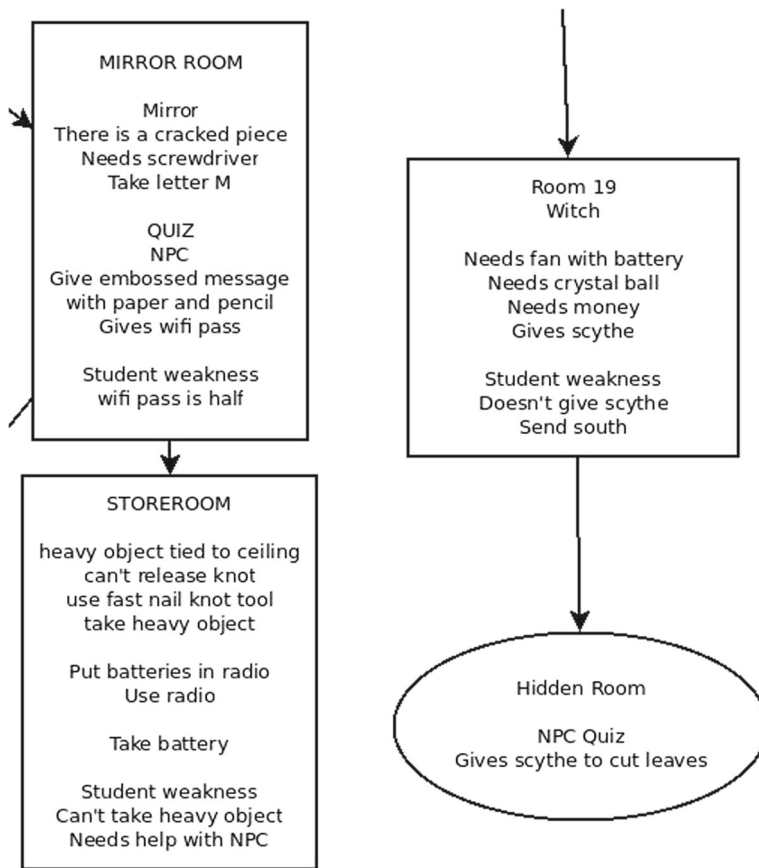


Fig. 6 New room (ellipse) added to the game

such as in which room s/he enters and what is inside the room. The educational part contains information on how fuzzy logic influences the choice of quiz questions. Finally, the third and final part reports the player's success rate, how s/he is ranked, and how the game itself is affected.

5.1 Example 1

5.1.1 Interaction 1

Description

George entered a room where an NPC character wearing a tie stands (Fig. 7). The character offered to help the hero provided that he (the hero) successfully answers several questions about HTML.

Educational part

George was invited for the first time to solve a quiz, so his knowledge level was "Unknown" ($\mu_{Un}(x) = 1, \mu_{UK}(x) = 0, \mu_K(x) = 0, \mu_L(x) = 0$). Under the rule that

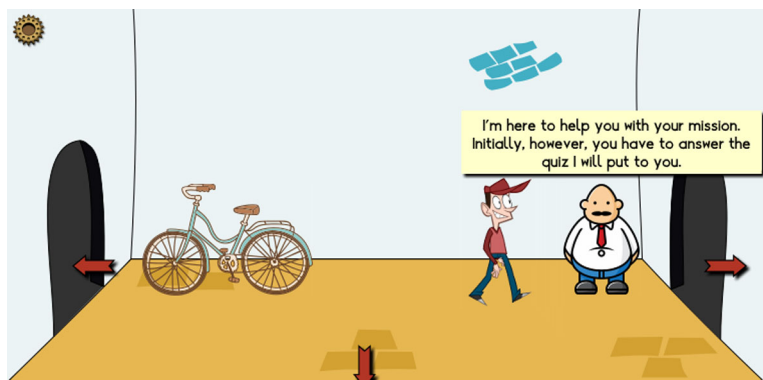


Fig. 7 NPC offers to help

applies to the first quiz (Table 1, rule 1), the player had to answer 15 random easy questions (Fig. 8).

Game part

George’s success rate was 93%. Thus, according to the membership functions (1), (2), (3) and (4), the values of fuzzy sets became $\mu_{U_n}(x) = 0$, $\mu_{U_K}(x) = 0$, $\mu_K(x) = 0$, $\mu_L(x) = 1$. So, George’s cognitive state changed and he was belonging 100% to the “Learned” fuzzy set. Regarding the adaptivity in the game context and based on the game rules, the player successfully solved the quiz. The character rewarded him by giving a can full of oil to the hero (Fig. 9). The oil-can was essential for the continuity of the game as the hero should give it to a child to put oil on his motorway board game.

5.1.2 Interaction 2

Description

Afterwards, George went to a room where a heavy object was attached to the ceiling (Fig. 10). An NPC character prevented the hero from cutting the rope and lowering the

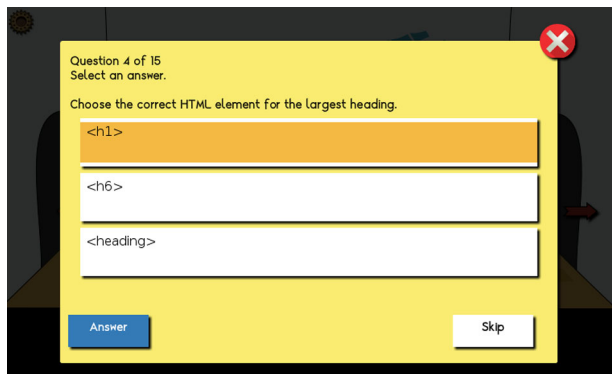


Fig. 8 1st quiz contains 15 easy questions

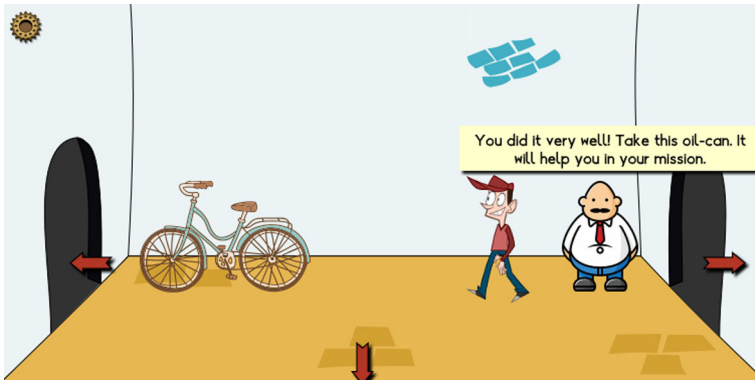


Fig. 9 NPC gives a can full of oil as a reward

object. However, the character offered to leave the room in case the hero answered the quiz that assigns to him.

Educational part

It was the second quiz the player's knowledge level was 100 "Learned" ($\mu_{Un}(x) = 0$, $\mu_{UK}(x) = 0$, $\mu_K(x) = 0$, $\mu_L(x) = 1$). So according to rule 6, the player had to answer 15 hard questions (Fig. 11).

Game part

The player answered correctly in all the questions. Therefore, from the membership functions (1), (2), (3) and (4), it appears that his classification in the "Learned" fuzzy set remained the same. Therefore, the NPC character leaved the room, and George may cut the rope and pick up the object (Fig. 12).

5.1.3 Interaction 3

Description



Fig. 10 Room with a heavy object

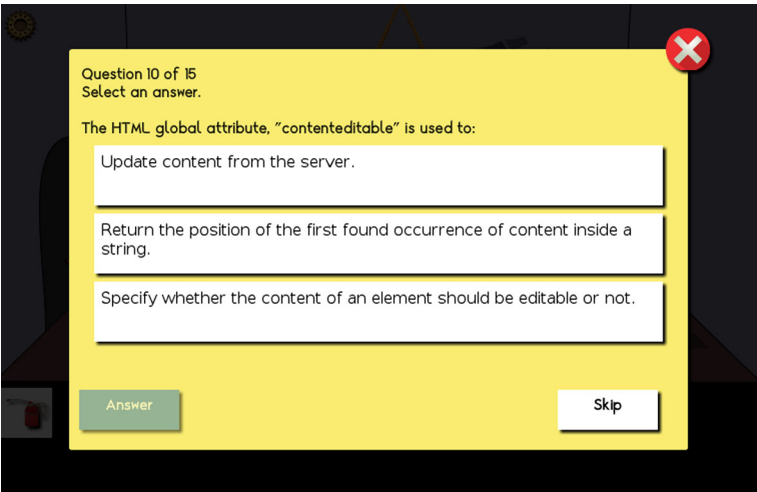


Fig. 11 Quiz consisted of 15 hard questions

George entered a room where a witch is. After giving her the objects she asked (crystal ball, money, fan), he had to answer the quiz that the witch assigned to him (Fig. 13).

Educational part

The last two rankings of George were “Learned”, so the quiz consisted of 15 random, hard questions (Table 1, rule 19).

Game part

After answering the quiz, the success rate of George was 96%. Therefore, he remained in the “Learned” fuzzy set with $\mu_L(x) = 1$. The player resolved three quizzes with ranking “Learned” and continued the game where he still had a final quiz to complete the game (Fig. 14). We observe that the game requires excellent player performance in multiple quizzes to establish the view that the player knows the learning subject.



Fig. 12 NPC leaves and hero can pick up the object



Fig. 13 The witch asks questions about HTML

5.2 Example 2

5.2.1 Interaction 1

Description

Nancy entered the room with the heavy object, where the NPC character was there to prevent her from getting the object. It was the first quiz, so Nancy was asked to answer 15 random, easy questions.

Game part

Nancy's success rate was approximately 53%. Based on the membership functions, she was classified 100% to the "Unsatisfactorily Known" set ($\mu_{U_n}(x) = 0, \mu_{U_K}(x) = 1, \mu_K(x) = 0, \mu_L(x) = 0$). Rule 1 of game context applied, so Nancy didn't get any help (Fig. 15).



Fig. 14 Final test appears after several rankings in "Learned" fuzzy set



Fig. 15 Hero didn’t get any help due to “Unsatisfactory Known” set

5.2.2 Interaction 2

Description

Nancy didn’t stay in the same room to try to resolve a new quiz, but went to the room with the NPC character wearing a tie. The NPC raised, in turn, a quiz with questions.

Educational part

It was the second interaction, and Nancy’s model was already to the “Unsatisfactorily Known” set. Rule number 3 of Table 1 applied, according to which the quiz consisted of 7 easy questions and 8 moderate difficulty questions (Fig. 16).

Game part

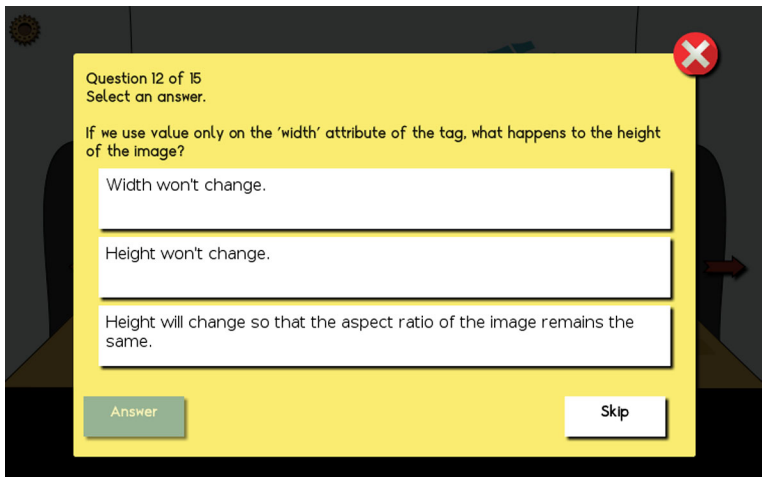


Fig. 16 The quiz consists of 7 easy questions and 8 moderate difficulty questions

Nancy had a success rate equal to 93%. Therefore, she was ranked in the “Learned” set with a 100% participation ($\mu_{Un}(x) = 0, \mu_{UK}(x) = 0, \mu_K(x) = 0, \mu_L(x) = 1$). The NPC character wearing the tie gave the can full of oil to Nancy.

5.2.3 Interaction 3

Description

Nancy returned to the room with the heavy object to try again to solve the quiz.

Educational part

It was the third interaction. The two previous rankings were “Unsatisfactorily Known” and “Learned”. Therefore, according to rule 16 in Table 1, the quiz consisted of 7 easy questions and 8 moderate difficulty questions.

Game part

Nancy had a 100% success rate and she was ranked to “Learned” set ($\mu_{Un}(x) = 0, \mu_{UK}(x) = 0, \mu_K(x) = 0, \mu_L(x) = 1$). The character standing there left and Nancy picked up the object. Afterwards, she continued playing and tried to resolve the next quiz presented along her way.

5.3 Example 3

5.3.1 Interaction 1

Description

Dimitra entered the room with the character wearing a tie, who started asking questions about HTML.

Educational part

It was the first interaction, so the quiz consisted of 15 random, easy questions (Table 1, rule 1).

Game part

Dimitra had a success rate of 80%. She was ranked 100% in the “Known” set ($\mu_{Un}(x) = 0, \mu_{UK}(x) = 0, \mu_K(x) = 1, \mu_L(x) = 0$), so rule number 3 (primary quiz, set “Known” but not “Learned”) was applied. Hence, Dimitra’s performance was satisfactory but had minor weaknesses. Subsequently, the NPC character offered a reward which was lower than in the case of classification to “Learned” set. More particularly, he gave an empty can, and Dimitra had to find oil elsewhere to fill the can (Fig. 17). A diagram of possible game states interconnections between them is represented in Fig. 18. At the same time, a process expanding the game started in the background. The expansion of the game was perceived in the next interaction.

5.3.2 Interaction 2

Description

Dimitra went to the kitchen where initially there was no NPC character. Due to the previous interaction with the character with the tie, there was now a chef in the kitchen who set a new quiz to the player (Fig. 19). So a new character appeared, who assigned a repeat test to Dimitra to help her overcome her weaknesses.

Educational part

It was the second interaction, so according to rule 5 in Table 1, the quiz consisted of 7 moderate difficulty questions and 8 hard questions.

Game part

Dimitra answered correctly with a success rate of 93%), and therefore was classified to the “Learned” fuzzy set by 100% ($\mu_{Un}(x) = 0$, $\mu_{UK}(x) = 0$, $\mu_K(x) = 0$, $\mu_L(x) = 1$). The chef indicated a secret door, and Dimitra entered a new secret room, which did not exist before (Figs. 20 and 21). This room contained a container of oil, from where Dimitra could fill the empty can and continue the adventure. The quiz was additional one and did not belong to the original that Dimitra should solve mandatory. However, due to the small difficulties encountered in the previous interaction, this quiz was produced and must be resolved. In this case, successful completion of the quiz was valid only if Dimitra was ranked in the “Learned” set, even partially.

5.3.3 Interaction 3

Description

Dimitra went to the room with the heavy object.

Educational part

It was the third interaction, therefore the last two rankings, i.e. 100% “Learned” and 100% “Known” respectively, were taken into account. So, according to rule 18 in Table 1, the quiz consisted of 7 moderate difficulty questions and 8 hard questions.

Game part

Dimitra answered with a success rate approximately equal to 67%. Therefore, according to the membership functions, it was 60% of the total “Unsatisfactorily Known” and 40% of

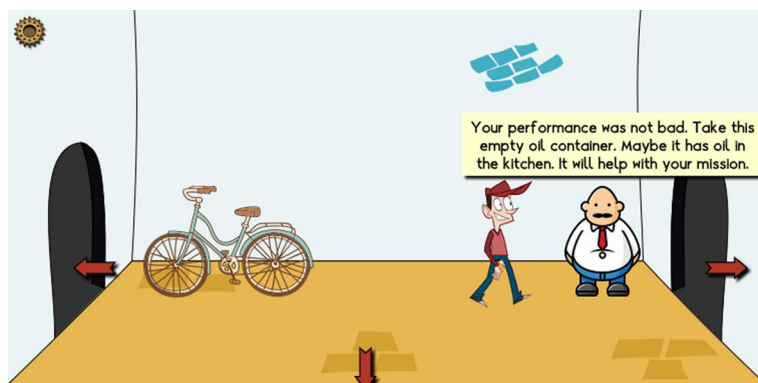


Fig. 17 Because of “Known” set, the oil-can the player earns, is empty

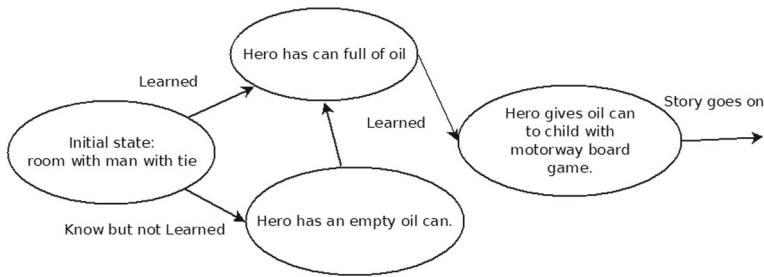


Fig. 18 Diagram of possible game states in the room with the man wearing a tie

the total “Known” ($\mu_{Un}(x) = 1, \mu_{UK}(x) = 0.6, \mu_K(x) = 0.4, \mu_L(x) = 0$). Because the quiz was in the category of primary ones, rule 3 applied. According to this rule, Dimitra earned a little reward and continued her adventure. The NPC character lying there left the room, but Dimitra discovered that the object was so heavy that the hero could not lift it (Fig. 22). The game expanded silently, and Dimitra had to find a way to help the hero get the item.

5.3.4 Interaction 4

Description

Dimitra went to the room where the NPC character with the tie was standing. In the room, there was now a new NPC character, a muscular athlete, who assigned a new quiz to Dimitra (Fig. 23).

Educational part

The last two rankings of Dimitra’s knowledge level defined the level of difficulty of the questions. According to Table 1, when the last rank is “Known” and the penultimate is “Learned”, the quiz consists of 15 hard questions (rule 15).

Game part

Dimitra answered correctly to all questions (100% success rate) and was ranked again in the “Learned” fuzzy set with a 100% grade ($\mu_{Un}(x) = 0, \mu_{UK}(x) = 0, \mu_K(x) = 0, \mu_L(x) = 1$). The strong character followed the hero and helped him lift the heavy object.



Fig. 19 The chef appears only if the player has weaknesses

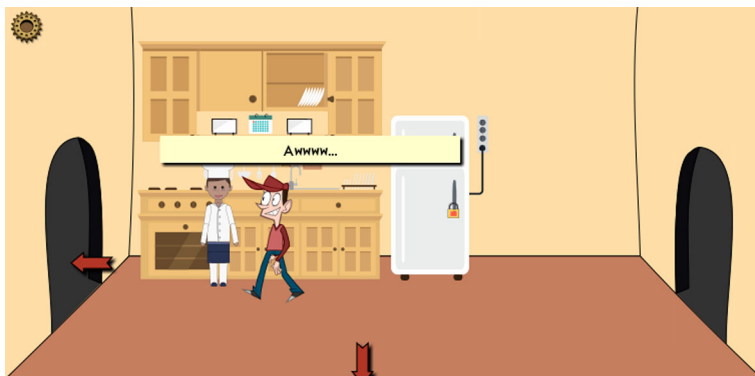


Fig. 20 A door to a new room appears

Afterwards, they placed the heavy object on a platform to open a hidden hatch. In case the quiz was additional, and Dimitra answered correctly to 80% of the questions, she would be classified completely to fuzzy set “Known” ($\mu_{U_n}(x) = 0, \mu_{U_K}(x) = 0, \mu_K(x) = 1, \mu_L(x) = 0$), and the strong character would not give her any reward but would assign to her a repeat test.

The above are indicative examples of how the scenario can be expanded with regard to student modeling. Various combinations of the above examples may take place, which can result in an even greater expansion scenario incorporating new locations, items, and NPC characters. Whenever the player is ranked to “Known” set, but not to “Learned” set in any of the basic quizzes, the game will expand accordingly, and the player will have to resolve new quizzes and new puzzles. The implicit extension of the scenario aims at further practice in the cognitive subject regarding educational aspect and increasing motivation and engagement regarding the game side.

In addition, if the player fails in consecutive quizzes, the educational component sends to the game information regarding the player’s most recent mistakes. Subsequently, an agent may appear in a room to suggest to the player which part of the curriculum can help her/him to find the answer to one of her/his mistakes (Fig. 24). Alternatively, the hero may find on the floor a piece of paper that will include hint with a link to the corresponding syllabus.



Fig. 21 There is a container full of oil in the new room



Fig. 22 The player has minor weaknesses, hence the object is too heavy

6 Experimental data

In this section, we present some experimental data that have been derived from the use of the presented game in real-life conditions. As we mentioned above, a group of 60 users, which were learners of a postgraduate program in the field of Informatics at the University of Piraeus, used the presented educational game. All learners attended lessons of HTML in class for a period of one month before playing the game and then they played the game in a period of two weeks. The average number of quizzes, which they were asked to answer in order to complete the game and be considered that they have learned the HTML programming language, was 7.88%. In more details, 28.33% of the users needed to answer to a maximum number of 5 quizzes; 58.33% of the users were asked to answer from 6 up to 10 quizzes; and 13.33% of the users answered more than 10 quizzes. Furthermore, the average difficulty level of the questions that were asked to be answered by the 60 learners was: 22.29% easy, 28.61% moderate and 49.10% hard. These experimental data are depicted in Table 2.

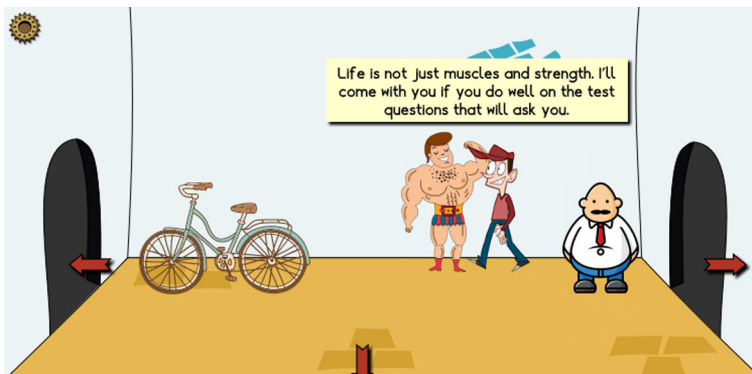


Fig. 23 A muscle man can help the player to lift the heavy object

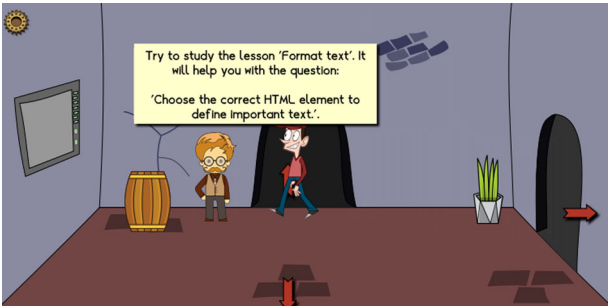


Fig. 24 An agent provides a hint to one of the latest questions answered wrong

Furthermore, in Table 3 we present a sample of the alterations of the values of the fuzzy sets that show the knowledge level of the learners during their interactions with the game.

7 Evaluation

The phase of evaluation is crucial for any application since it underlines the success of it. So, in this section the process and results of the evaluation of the presented game were described. There are several techniques of evaluation; however, there is not either a standard agreed on an evaluation technique or framework for games evaluation. The most common evaluation methods are questionnaires and experiments. However, according to Petri and Gresse von Wangenheim [18] questionnaires consist of the most commonly used method for games evaluation. The evaluation of games and even more the evaluation of educational games is a complex process. In the case of the educational game either the game (interface, usefulness, plot) or the learning effectiveness (the acquisition of knowledge, motivating, engaging) has to be assessed. Notably, the educational effectiveness that examines if the game is motivating and educationally beneficial to students has to be evaluated, since even the motivational advantage of educational games may be questioned as in a classroom, there may be students who do not like games or students who find it hard to navigate through the virtual world and thus may not be able to benefit to the full from the educational content of the software [24].

Furthermore, usability and likeability of the game have to be accessed [25]. The evaluation of these two factors is crucial since modern computer games usually have complex user interfaces, to be more attractive, but this can lead to a not functional game. On the other

Table 2 Experimental data

Number of interactions - quizzes (q)	Average numbers of learners - users	Average difficulty level of the questions of all quizzes		
		Easy	Moderate	Hard
$q \leq 5$	28.33%	20.36%	21.52%	58.12%
$6 \leq q \leq 10$	58.33%	22.08%	31.53%	46.39%
$q > 10$	13.33%	24.45%	32.78%	42.78%
Totally		22.29%	28.61%	49.10%

Table 3 A sample of the alterations of the fuzzy sets values

Users	Values of fuzzy sets per interaction ($\mu_{Un}, \mu_{UK}, \mu_K, \mu_L$)						Total number of questions				Number of quizzes
	Inter. 1	Inter. 2	Inter. 3	...	Inter. n-1	Inter. n	Easy	Moderate	Hard	Sum	
1	(1, 0, 0, 0)	(0, 0, 1, 0)	(0, 0, 4, 0, 6, 0)	...	(0, 0, 0, 1)	(0, 0, 0, 1)	15	29	60	104	6
2	(1, 0, 0, 0)	(0, 0, 1, 0)	(0, 0, 4, 0, 6, 0)	...	(0, 0, 0, 8, 0, 2)	(0, 0, 0, 1)	15	29	45	89	5
3	(1, 0, 0, 0)	(0, 1, 0, 0)	(0, 0, 1, 0)	...	(0, 0, 0, 4, 0, 6)	(0, 0, 0, 1)	29	37	54	120	8
4	(1, 0, 0, 0)	(0, 0, 1, 0)	(0, 0, 1, 0)	...	(0, 0, 0, 8, 0, 2)	(0, 0, 0, 1)	15	14	46	75	5
5	(1, 0, 0, 0)	(0, 0, 1, 0)	(0, 0, 6, 0, 4, 0)	...	(0, 0, 0, 1)	(0, 0, 0, 1)	15	7	38	60	4
6	(1, 0, 0, 0)	(0, 1, 0, 0)	(0, 1 0, 0)	...	(0, 0, 0, 1)	(0, 0, 0, 1)	29	44	77	150	10
7	(1, 0, 0, 0)	(1, 0, 0, 0)	(0, 2, 0, 8, 0, 0)	...	(0, 0, 0, 6, 0, 4)	(0, 0, 0, 8, 0, 2)	52	52	45	149	10
8	(1, 0, 0, 0)	(0, 4, 0, 6, 0, 0)	(0, 1, 0, 0)	...	(0, 0, 0, 6, 0, 4)	(0, 0, 0, 1)	44	59	77	180	12
9	(1, 0, 0, 0)	(0, 0, 1, 0)	(0, 0, 1, 0)	...	(0, 0, 0, 4, 0, 6)	(0, 0, 0, 2, 0, 8)	29	29	46	104	7
10	(1, 0, 0, 0)	(0, 0, 8, 0, 2, 0)	(0, 0, 1, 0)	...	(0, 0, 0, 6, 0, 4)	(0, 0, 0, 1)	15	59	61	135	9

Table 4 The questionnaire

Questions			Answers (circle one for each question) 1: unsatisfactory, 2: Improvement needed, 3: meet expectations, 4: exceeds expectations, 5: excellent				
Usability	Q1	Is the game's goal clear?	1	2	3	4	5
	Q2	Are the game's rules clear?	1	2	3	4	5
	Q3	How easy is it to navigate to the game?	1	2	3	4	5
Likeability	Q4	How easy is to play the game?	1	2	3	4	5
	Q5	How interesting is the game?	1	2	3	4	5
	Q6	How much did you like the graphics / illustrations of the game?	1	2	3	4	5
	Q7	Rate the quality of the game's multimedia elements.	1	2	3	4	5
	Q8	How much did you like the particular game?	1	2	3	4	5
Educational effectiveness	Q9	The game design is attractive?	1	2	3	4	5
	Q10	Do you have fun with the game?	1	2	3	4	5
	Q11	Does gameplay motivate you to participate in it?	1	2	3	4	5
	Q12	Rate your involvement in the game?	1	2	3	4	5
	Q13	Is the learning goal clear?	1	2	3	4	5
	Q14	Do you like to use the particular game for learning the knowledge domain about HTML programming?	1	2	3	4	5
	Q15	Would you like to use the particular game as a tool for supporting your learning at other domain concepts?	1	2	3	4	5
	Q16	How interesting is the learning context (quizzes) of the game?	1	2	3	4	5
	Q17	Does the game help you to obtain new knowledge about HTML programming?	1	2	3	4	5
	Q18	Did you feel confident that you were learning, passing through the game?	1	2	3	4	5
Adaptivity	Q19	The game's plot and the continuous quizzes are interesting?	1	2	3	4	5
	Q20	Rate the game's quizzes relevance to your knowledge level.	1	2	3	4	5
	Q21	Do you think that the game identified each time your knowledge level?	1	2	3	4	5

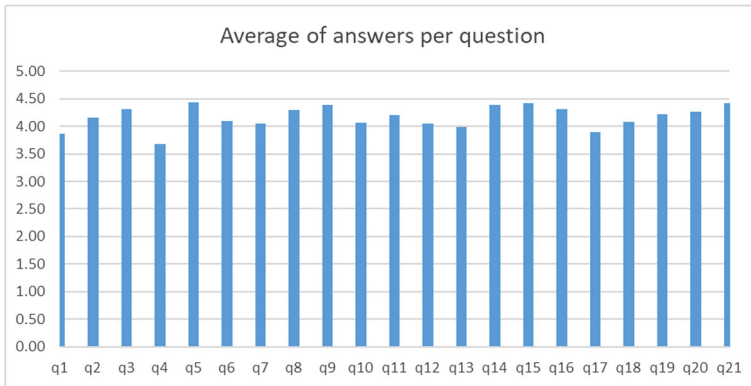


Fig. 25 The results of the game's evaluation

hand, a computer game that incorporates educational content may lose its attractiveness and appeal. Also, the factor of adaptivity has to be accessed. In other words, it is crucial to evaluate if the educational content of the game (quizzes) adapts to the knowledge level and the needs of the learners.

Taking into consideration the above, a questionnaire has been conducted to evaluate the presented educational game. Through the particular questionnaire the following four factors are accessed: i) educational effectiveness, ii) usability, iii) likeability, and iv) adaptivity. In particular, the questionnaire, which has been conducted for the game's evaluation, consists of 21 questions that are close-ended based on a Likert scale with the responses ranging

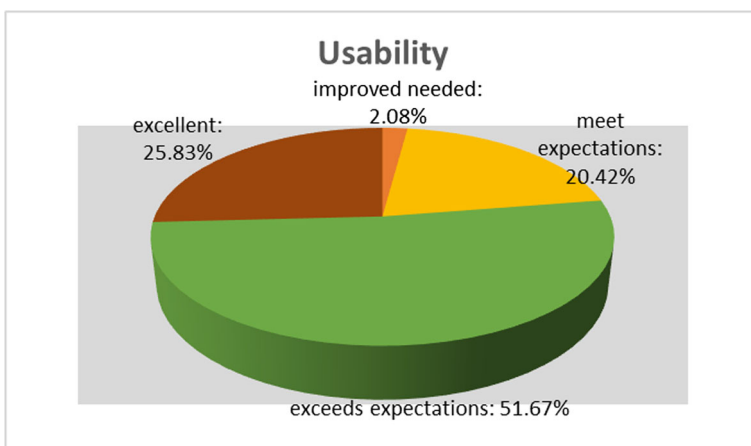


Fig. 26 Usability results

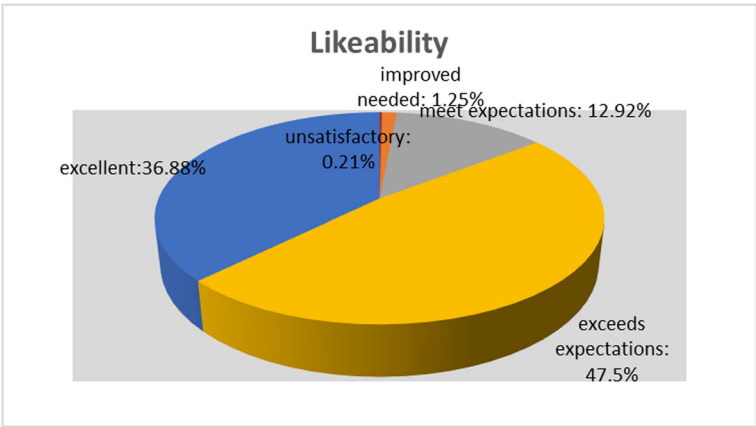


Fig. 27 Likeability results

from “Unsatisfactory” (1) to “Excellent” (5). 6 questions concerning the educational effectiveness, 4 questions concern usability, 8 questions concern likeability, and 3 questions concerning the game’s adaptivity. The questionnaire is depicted in Table 4.

A group of 60 users of different ages used the presented game. All the users were learners of a postgraduate program in the field of informatics at the University of Piraeus. Learners attended lessons of HTML in class for one month before playing the game. Then, they played the game in a period of three weeks. After this period, they were asked to answered the questionnaire below. The results of the evaluation are depicted in Figs. 25, 26, 27, 28 and 29, and they are very positive and encouraging.

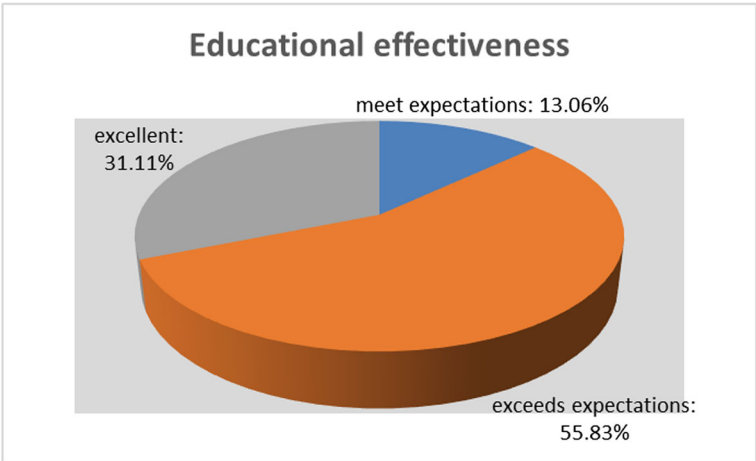


Fig. 28 Educational effectiveness results

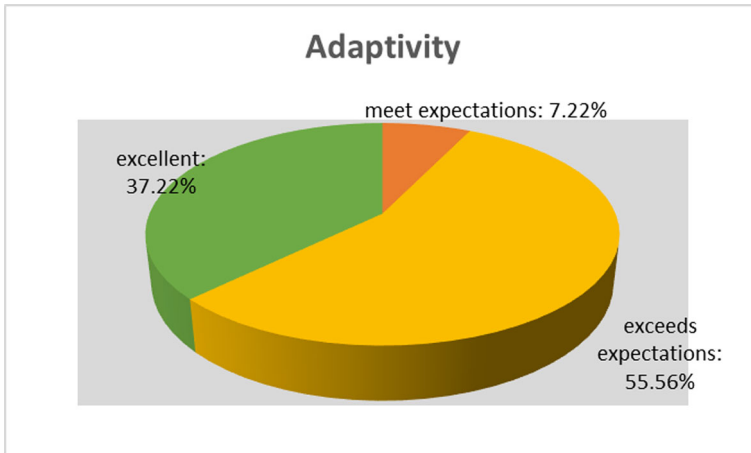


Fig. 29 Adaptivity results

8 Conclusions

Computer games incorporate rich content such as graphics, animation, audio, and video, making them particularly appealing. Because of their entertaining nature can be used successfully in the learning process of a cognitive field. In particular, adventure computer games are based primarily on narrative rather than action and are thus considered profitable in use in education. Also, it is essential that an educational application, whether it is an educational game or not, can be personalised to the needs of each user separately.

In this article, we presented an educational adventure game that aims to teach users HTML programming. The hero wanders in different rooms, collects items and converses with characters. The game incorporates adaptive elements to provide the user with a personalised learning experience based on the learner's level of knowledge. Fuzzy logic algorithms calculate the latter because it is a concept that cannot be quantified or described with a discrete value. Adaptivity is applied to both the educational component of the system and the game itself. In the first case, it affects the level of difficulty of the questions to help the user to overcome any weaknesses and improve. The second part extends the game with new puzzles, new rooms, and tests to increase the motivation of the player and allow her/him to take part in further quizzes for practice.

During the research, we used the questionnaire survey method to collect feedback from the students. The questions were categorized into the concepts of usability, likeability, educational effectiveness, adaptivity to evaluate both the game and its educational part. The results were positive and encouraging concerning either the game or the learning concepts. Students perceived the game pleasantly while the use of fuzzy logic techniques seemed to approach to a great extent the user's knowledge level. The adaptivity of quiz questions matched users' needs and helped them overcome any difficulties. Furthermore, the expansion of the game scenario kept some users engaged longer. The majority of users stated the desire to use the game in other educational subjects as well.

The contribution of this research is very important in the area of educational software. The reason for this is the fact that the presented work provides a novel mechanism that uses fuzzy logic to build student models and adapt an educational game on the fly. Hence, the presented educational game reacts to particular and individual educational needs of students

by adapting the quizzes of the game and by altering the scenario of the game and the plot of the story at run time. As a result, students may gain further training without losing the motivation of the surprise of the game.

The findings of our study show that an educational game that combines adventure games' elements, intelligence through fuzzy logic and adaptivity can be very beneficial in the learning process. In the future, we plan to take advantage of the game in learning additional cognitive fields. We have already implemented an authoring tool that supports the entry of new courses in the game either by the administrator or by a teacher. We aim to offer the authoring tool to teachers to import educational material and monitor the performance of their students.

References

- Boyle T (1997) Design for multimedia learning. Prentice-Hall Inc., Englewood Cliffs
- Chrysafiadi K, Virvou M (2010) Modeling student's knowledge on programming using fuzzy techniques. In: Intelligent Interactive Multimedia Systems and Services. Springer, pp 23–32
- Chrysafiadi K, Virvou M et al (2016) Advances in personalized web-based education. Springer, Berlin
- D'Apice C, Grieco C, Piscopo R, Liscio L (2015) Dms2015short-2: Advanced learning technologies for learning in the enterprise: Design of an educational adventure game to teach computer security. *J Vis Lang Comput* 31:260–266
- Drigas AS, Argyri K, Vrettaros J (2009) Decade review (1999–2009): artificial intelligence techniques in student modeling. In: World Summit on Knowledge Society. Springer, pp 552–564
- ESA (2014) Esa 2014 annual report. <http://www.theesa.com/wp-content/uploads/2015/03/ESA-2014-Annual-Report.pdf>, accessed: 2018-04-10
- Gaeta M, Loia V, Mangione GR, Orciuoli F, Ritrovato P, Salerno S (2014) A methodology and an authoring tool for creating complex learning objects to support interactive storytelling. *Comput Hum Behav* 31:620–637
- Hou HT, Li MC (2014) Evaluating multiple aspects of a digital educational problem-solving-based adventure game. *Comput Hum Behav* 30:29–38
- Jeremić Z, Jovanović J, Gašević D (2012) Student modeling and assessment in intelligent tutoring of software patterns. *Expert Syst Appl* 39(1):210–222
- Karpinskyj S, Zambetta F, Cavedon L (2014) Video game personalisation techniques: a comprehensive survey. *Entertain Comput* 5(4):211–218
- Lo JJ, Chan YC, Yeh SW (2012) Designing an adaptive web-based learning system based on students cognitive styles identified online. *Comput Educ* 58(1):209–222
- Martí-Parreño J, Galbis-Córdova A, Miquel-Romero MJ (2017) Students' attitude towards the use of educational video games to develop competencies. *Computers in Human Behavior*
- Mehm F, Göbel S, Steinmetz R (2012) Authoring of serious adventure games in storytec. In: E-learning and games for training, education, Health and Sports. Springer, pp 144–154
- Minović M, Milovanović M, Šošević U, Mác González (2015) Visualisation of student learning model in serious games. *Comput Hum Behav* 47:98–107
- Moreno-Ger P, Sierra JL, Martínez-Ortiz I, Fernández-Manjón B (2007) A documental approach to adventure game development. *Sci Comput Programm* 67(1):3–31
- Papert S (1993) The children's machine: Rethinking school in the age of the computer. ERIC
- Peirce N, Conlan O, Wade V (2008) Adaptive educational games: Providing non-invasive personalised learning experiences. In: 2008 Second IEEE International Conference on Digital games and intelligent toys based education. IEEE, pp 28–35
- Petri G, von Wangenheim CG (2017) How games for computing education are evaluated? a systematic literature review. *Comput Educ* 107:68–90
- Pilegard C, Mayer RE (2016) Improving academic learning from computer-based narrative games. *Contemp Educ Psychol* 44:12–20
- Pivec M (2007) Play and learn: potentials of game-based learning. *Br J Educ Technol* 38(3):387–393
- Polycarpou I, Krausea J, Rader C, Kembel C, Poupore C, Chiu E (2010) Math-city: an educational game for k-12 mathematics. *Procedia-Soc Behav Sci* 9:845–850
- Riemer V, Schrader C (2015) Learning with quizzes, simulations, and adventures: Students' attitudes, perceptions and intentions to learn with different types of serious games. *Comput Educ* 88:160–168

23. Papadimitriou S, Virvou M (2017) Adaptivity in scenarios in an educational adventure game. In: 2017 8th International Conference on Information, Intelligence, Systems & Applications (IISA). IEEE, pp 1–6
24. Virvou M, Katsionis G, Manos K (2005) Combining software games with education: Evaluation of its educational effectiveness. *J Educ Technol Soc* 8(2):54–65
25. Virvou M, Katsionis G (2008) On the usability and likeability of virtual reality games for education: The case of vr-engage. *Comput Educ* 50(1):154–178
26. Wikipedia (2018a) List of video game genres. https://en.wikipedia.org/wiki/List_of_video_game_genres, accessed: 2018-04-10
27. Wikipedia (2018b) Colossal cave adventure. https://en.wikipedia.org/wiki/Colossal_Cave_Adventure, accessed: 2018-04-10
28. Zadeh LA (1996) Fuzzy logic= computing with words. *IEEE Trans Fuzzy Syst* 4(2):103–111
29. Zapata-Rivera D (2010) Adaptive, assessment-based educational games. In: International Conference on Intelligent Tutoring Systems. Springer, pp 435–437

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